

Multichannel modulation

子通道频率 $f_n = \frac{n}{T}$, $n = 1, 2, \dots, N$

$\Rightarrow \Delta = \frac{1}{T}$ (剛好可以讓子載波正交)

Passband basis function of sub-channel :

$$\begin{cases} \phi(t) \cos(2\pi f_n t) \\ \phi(t) \sin(2\pi f_n t) \end{cases}, \text{ where } \phi(t) = \sqrt{\frac{2}{T}} \text{sinc} \frac{t}{T} \text{ is low-pass pulse.}$$

$$\Rightarrow \Phi(f) = \sqrt{2T} \text{rect}(Tf).$$

I/Q orthogonality :

$$\begin{aligned} \int_{-\infty}^{\infty} \phi(t) \cos(2\pi f_n t) \phi(t) \sin(2\pi f_n t) dt &= \frac{1}{2} \int_{-\infty}^{\infty} \phi^2(t) \sin(4\pi f_n t) dt \\ &= 0 \quad (\sin(4\pi f_n t) \text{ is odd and } \phi^2(t) \text{ is even}) \end{aligned}$$

Complex form :

$$r_n(t) = \frac{1}{\sqrt{2}} \phi(t) e^{j2\pi f_n t}$$

$$\psi_n(f) = \sqrt{T} \text{rect}(T(f-f_n))$$

Orthogonality of carriers of sub-channel :

$$\begin{aligned} \int_{-\infty}^{\infty} x(t) \cdot y^*(t) dt &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2}} \phi(t) e^{j2\pi f_n t} \left(\frac{1}{\sqrt{2}} \phi(t) e^{j2\pi f_k t} \right)^* dt \\ &= \int_{-\infty}^{\infty} \sqrt{T} \text{rect}(T(f-f_n)) \cdot \sqrt{T} \text{rect}(T(f-f_k)) df \\ &= \begin{cases} 1, & k = n \\ 0, & k \neq n \end{cases} \end{aligned}$$

$$\int_{-\frac{1}{2T}}^{\frac{1}{2T}} T \text{rect}(\cdot) \text{rect}(\cdot) df.$$

QAM signal model :

$$A_n = a_n + j \cdot b_n$$

$$\begin{aligned} S_n(t) &= a_n \phi(t) \cos(2\pi f_n t) - b_n \phi(t) \sin(2\pi f_n t) \\ &= \text{Re} \{ A_n \phi(t) e^{j2\pi f_n t} \}. \quad (\text{每個子通道都像一個 QAM}) \end{aligned}$$

$$r_n(t) = \frac{1}{\sqrt{2}} \phi(t) e^{j2\pi f_n t}$$

$$\Rightarrow \phi(t) e^{j2\pi f_n t} = \sqrt{2} r_n(t)$$

$$\begin{aligned} S_n(t) &= \text{Re} \{ A_n \sqrt{2} r_n(t) \} \\ &= \frac{\sqrt{2} A_n r_n(t) + (\sqrt{2} A_n r_n(t))^*}{2} \end{aligned}$$

$$\text{Then, } S_n(f) = \frac{\sqrt{2T}}{2} (A_n \text{rect}(T(f-f_n)) + A_n^* \text{rect}(T(f+f_n))).$$

$$S(t) = \sum_{n=1}^N S_n(t). \quad (\text{multichannel signal 是 } N \text{ 個 QAM 疊加})$$

Received signal :

$$x(t) = h(t) * s(t) + w(t)$$

$$X(f) = h(f) \cdot S(f) + W(f).$$

$$\text{If } N \rightarrow \infty, \quad H(f) \doteq \sum_{n=1}^N H(f_n) \text{rect}(T(f-f_n)) + H^*(f_n) \text{rect}(T(f+f_n)).$$

(N 很大, sub-channel 可視為 constant, 所以 frequency-selective channel $\rightarrow$ 多個 flat-fading subchannels.)

$$\sqrt{\frac{2}{T}} \text{sinc}\left(\frac{t}{T}\right) \xleftrightarrow{\text{F.T.}} \sqrt{2T} \text{rect}(Tf)$$

$$S(f) = \frac{\sqrt{2T}}{2} \sum_{n=1}^N A_n \text{rect}(T(f-f_n)) + A_n^* \text{rect}(T(f+f_n))$$

$$H(f) = \sum_{n=1}^N H(f_n) \text{rect}(T(f-f_n)) + H^*(f_n) \text{rect}(T(f+f_n)).$$

★ Channel output :

$$H(f) \cdot S(f) \triangleq \sqrt{\frac{T}{2}} \sum_{n=1}^N [A_n H(f_n) \text{rect}(T(f-f_n)) + A_n^* H^*(f_n) \text{rect}(T(f+f_n))] \quad \begin{matrix} \sqrt{\frac{T}{2}} \text{rect}(T(f-f_n)) \\ \uparrow \\ \frac{1}{\sqrt{T}} \text{sinc}(\frac{t}{T}) e^{j2\pi f_n t} \\ = \frac{1}{2} \phi_n(t) e^{j2\pi f_n t} \end{matrix}$$

$$\Rightarrow h(t) * s(t) = \sum_{n=1}^N \text{Re} \{ A_n H(f_n) \phi_n(t) e^{j2\pi f_n t} \} = \sum_{n=1}^N \text{Re} \{ \sqrt{2} A_n H(f_n) \psi_n(t) \}$$

$$\Rightarrow x(t) = \sum_{n=1}^N \text{Re} \{ \sqrt{2} A_n H(f_n) \psi_n(t) \} + w(t)$$

A_n 經過通過後又被乘上 $H(f_n)$

$\Rightarrow$ 子通道模型 : Transmitted symbol $\rightarrow x H(f_n) \rightarrow$ add noise

$$\frac{1}{2} \phi_n(t) + \frac{1}{2} \phi_n^*(t) = \text{Re} \{ \psi_n(t) \}$$

$A_n \cdot H(f_n)$ is constant, $\phi_n(t) e^{j2\pi f_n t} = \sqrt{2} \psi_n(t)$

$\Rightarrow$ get the signal $s * h$

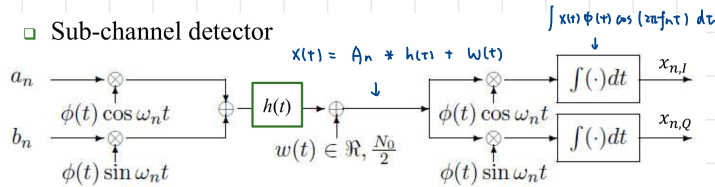
Subchannel detector :

$$W_n = w_{n,i} + j w_{n,q}, \text{ where } w_{n,i}, w_{n,q} \sim \mathcal{N}(0, \frac{N_0}{2})$$

$$\text{Let } X_n = x_{n,i} + j x_{n,q}$$

$$\text{Then, } X_n = A_n H(f_n) + W_n \quad (\text{子通道是 complex AWGN channel})$$

□ Sub-channel detector



$$x_{n,i} = \int_{-\infty}^{\infty} x(t) \phi(t) \cos(2\pi f_n t) dt = \text{Re} \{ A_n H(f_n) \} + w_{n,i}$$

$$x_{n,q} = \int_{-\infty}^{\infty} x(t) \phi(t) \sin(2\pi f_n t) dt = \text{Im} \{ A_n H(f_n) \} + w_{n,q}$$

ML detector :

$$\hat{A}_n = \frac{X_n}{H(f_n)} \text{ if we have } H(f_n), \forall n$$

(可直接 equalize)

$$\hat{A}_n = \underset{A \in C}{\text{argmin}} |X_n - A H(f_n)|^2, \quad C \text{ is constellation set}$$

SNR of sub-channel :

$$n\text{-th SNR} : \text{SNR}_n = \frac{|H(f_n)|^2 P_n}{\sigma_n^2}, \text{ where } \sigma_n^2 = N_0 \Delta f$$

P_n : Power of n -th sub-channel

$|H(f_n)|^2$: Channel power gain

σ_n^2 : Noise power

★ Overall SNR and Effective rate :

$$\text{Define } R = \frac{1}{N} \sum_{n=1}^N R_n \quad (R_n = \frac{1}{2} \log_2 (1 + \frac{\text{SNR}_n}{\Gamma}))$$

$$\text{In high SNR : } R = \frac{1}{2N} \sum_{n=1}^N \log_2 (1 + \frac{P_n}{\Gamma \sigma_n^2}) \quad \leftarrow \times |H(f_n)|^2 \text{ 更完整}$$

$$= \frac{1}{2} \log_2 \left[\prod_{n=1}^N (1 + \frac{P_n}{\Gamma \sigma_n^2}) \right]^{\frac{1}{N}} \triangleq \frac{1}{2} \log_2 (1 + \frac{\text{SNR}_{\text{overall}}}{\Gamma})$$

$$\Rightarrow \text{SNR}_{\text{overall}} = \Gamma \left[\left(\prod_{n=1}^N (1 + \frac{P_n}{\Gamma \sigma_n^2}) \right)^{\frac{1}{N}} - 1 \right] \triangleq \prod_{n=1}^N \left(\frac{P_n}{\sigma_n^2} \right)^{\frac{1}{N}}$$

★ Loading / Water Filling :

$$\text{Let } g_n = |H(f_n)|$$

$$\text{Goal : Maximize } R = \frac{1}{2N} \sum_{n=1}^N \log_2 (1 + \frac{g_n^2 P_n}{\Gamma \sigma_n^2}) \text{ with restriction } \sum_{n=1}^N P_n = P$$

$$\mathcal{J} = \frac{1}{2N} \sum_{n=1}^N \log_2 (1 + \frac{g_n^2 P_n}{\Gamma \sigma_n^2}) + \lambda (\sum_{n=1}^N P_n - P)$$

$$\frac{d}{dx} \log_2 u = \frac{d}{dx} \frac{\ln u}{\ln 2} = \frac{1}{\ln 2} \cdot \frac{u'}{u}$$

$$\frac{\partial \mathcal{J}}{\partial P_n} = \frac{1}{2N} \log_2 e \cdot \left(\frac{\frac{g_n^2}{\Gamma \sigma_n^2}}{1 + \frac{g_n^2 P_n}{\Gamma \sigma_n^2}} \right) + \lambda = 0$$

Let $0 = 0$, we get

$$\lambda = \frac{\frac{1}{2N} \log_2 e}{P_n + \frac{\Gamma \sigma_n^2}{g_n^2}} \triangleq K$$

Multiplicator $\lambda \in \mathbb{R}$ is fixed $\forall n$, $\frac{1}{2N} \log_2 e \in \mathbb{R}$

Thus, the denominator must be a constant K

★ Water filling .

$$\text{Since } K = P_n + \frac{\sigma_n^2}{g_n^2}, \quad P_n = K - \frac{\sigma_n^2}{g_n^2}.$$

$$\Rightarrow P_n = \max(0, K - \frac{\sigma_n^2}{g_n^2}).$$

K 可視為 water level, $P \uparrow$, $K \uparrow$

視為通道底部, g_n 大, 底部低, 拿更多功率.

DMT & OFDM

DMT signal : $s(t) = \sum_{k=0}^{N-1} S[k] e^{j2\pi f_k t} \text{rect}(\frac{t}{T})$, duration $T = nT_s$.

$$s[n] = s(nT_s) \text{ and } f_k = \frac{k}{NT_s}$$

$$\Rightarrow s[n] = \sum_{k=0}^{N-1} S[k] e^{j2\pi f_k nT_s} = \sum_{k=0}^{N-1} S[k] e^{j\frac{2\pi}{N} kn}, \quad n = 0, 1, \dots, N-1$$

IDFT form (frequency-domain symbols $\xrightarrow{\text{IDFT}}$ time-domain samples)

把 impulse response 限制在 $[0, T_b]$, 有限個 sample $h_0, h_1, \dots, h_v$

$$\text{sampling rate : } \frac{1}{T_s} \Rightarrow T_b = (1+\nu)T_s$$

$$\text{No noise or SNR} \rightarrow \infty, \quad x[n] = \sum_{i=0}^v h_i s[n-i]$$

如果用 linear convolution, 會出現非此 block 的 $s[-1], s[-2], \dots, s[-v]$, 造成 IBI.

Cyclic Prefix :

$$\text{Let } s[-i] = s[N-i], \quad i = 1, 2, \dots, v.$$

If $N_{cp} \geq v$, there is no IBI.

Circular Convolution :

$$x[n] = \sum_{i=0}^v h_i (s[n-i] \bmod N)$$

Circular Matrix :

$$H = \begin{pmatrix} h_0 & h_1 & h_2 & \dots & h_v & 0 & \dots & 0 & 0 \\ 0 & h_0 & h_1 & \dots & h_{v-1} & h_v & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h_v & 0 & 0 & \dots & 0 & 0 & \dots & h_{v-1} & h_v \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h_2 & h_1 & h_0 & \dots & 0 & 0 & \dots & h_0 & h_1 \\ h_1 & h_2 & h_3 & \dots & 0 & 0 & \dots & 0 & h_0 \end{pmatrix}$$

$$X = [x[N-1], x[N-2], \dots, x[0]]$$

Matrix form : $x = Hs + w$ (x, s, w are vectors)

$$\text{DFT : } \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] \exp(-j\frac{2\pi}{N} kn), \quad k = 0, 1, \dots, N-1.$$

$$\text{IDFT : } \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X[k] \exp(j\frac{2\pi}{N} kn), \quad k = 0, 1, \dots, N-1.$$

DFT Matrix :

$$Q = [q_{kn}], \quad q_{kn} = \frac{1}{\sqrt{N}} \exp(-j\frac{2\pi}{N} kn), \quad k, n = 0, 1, \dots, N-1.$$

Then, $X = Qx$ and $x = Q^H X$ with $QQ^H = I$.

Eigenvalue :

$$\lambda_k = \sqrt{N} \sum_{n=0}^{N-1} h_n q_{kn}^* = \sum_{n=0}^{N-1} h_n \exp(j\frac{2\pi}{N} kn).$$

Then, $H = Q^H \Lambda Q$, $\Lambda = \text{diag}\{\lambda_0, \lambda_1, \dots, \lambda_{N-1}\}$.

$$\Rightarrow x = Q^H \Lambda Qs + w.$$

$$\begin{aligned} \Rightarrow X &= Qx = QQ^H \Lambda Qs + Qw \\ &= QQ^H \Lambda QQ^H S + QQ^H W \\ &= \Lambda S + W \end{aligned}$$

$$s(t) = \sum_{k=0}^{N-1} S[k] e^{j2\pi f_k t}, \quad T = nT_s$$

$$s[n] = s(nT_s)$$

$$= \sum_{k=0}^{N-1} S[k] e^{j2\pi f_k nT_s}$$

$$\downarrow f_k = \frac{k}{NT_s}$$

$$= \sum_{k=0}^{N-1} S[k] e^{j\frac{2\pi}{N} kn}$$

Channel $h_0 \dots h_v$ and $T_b = (1+\nu)T_s$.

$$X[n] = \sum_{i=0}^v h_i s[n-i], \quad n = 0 \sim N-1.$$

If $N_{cp} \geq v$ and $s[-i] = s[N-i]$

$$\text{Then, } X[n] = \sum h_i (s[n-i] \bmod N)$$

$\downarrow$

$$X = [x[N-1], \dots, x[0]]^T$$

$$H = \begin{pmatrix} h_0 & h_1 & \dots & h_v \\ 0 & h_0 & \dots & h_{v-1} \\ \vdots & \vdots & \ddots & \vdots \\ h_v & 0 & \dots & h_{v-1} \end{pmatrix}$$

$$\text{Let } Q = [q_{kn}], \quad q_{kn} = \frac{1}{\sqrt{N}} \exp(-j\frac{2\pi}{N} kn)$$

Eigenvalue of H

$$\lambda_k = \sqrt{N} \sum h_n q_{kn}^* = \sum h_n \exp(j\frac{2\pi}{N} kn)$$

Then $H = Q^H \Lambda Q$, $\Lambda = \text{diag}\{\lambda_0, \dots, \lambda_{N-1}\}$

$$x = Hs + w$$

$$= Q^H \Lambda Qs + w$$

$$X = Qx$$

$$= QQ^H \Lambda Qs + Qw$$

$$= QQ^H \Lambda QQ^H S + W$$

$$= \Lambda S + W$$

$$\Rightarrow X[k] = \lambda_k S[k] + W[k].$$

Thus, $X[k] = \lambda_k S[k] + W[k]$, where $W[k] \sim N(0, N_0)$
 $\hookrightarrow$ complex

Frequency domain equalization:

$$\hat{S}[k] = \frac{X[k]}{\lambda_k}$$

DMT / OFDM system flow:

傳送端

binary data $\rightarrow$ demultiplexer $\rightarrow$ constellation encoder $\rightarrow$ IDFT $\rightarrow$ parallel-to-serial

接收端

$\rightarrow$ cyclic prefix $\rightarrow$ DAC $\rightarrow$ channel $\rightarrow$ ADC $\rightarrow$ remove CP $\rightarrow$ serial-to-parallel

$\rightarrow$ DFT $\rightarrow$ equalization $\rightarrow$ decoder $\rightarrow$ multiplexer $\rightarrow$ estimate original input.

Synchronization

QPSK transmitted signal:

$$S_i(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t + (2i-1)\frac{\pi}{4}\right), \quad i = 1, 2, 3, 4.$$

Orthonormal basis function:

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t)$$

$$\phi_2(t) = -\sqrt{\frac{2}{T}} \sin(2\pi f_c t)$$

Receive signal:

$$X(t) = S_i(t) + W(t).$$

$$X_1(t) = \int_0^T X(t) \cdot \phi_1(t) dt = S_{i1} + W_1$$

$$X_2(t) = \int_0^T X(t) \cdot \phi_2(t) dt = S_{i2} + W_2$$

If i -th symbol is sent, then

$$X_1 = \sqrt{E} \cos\left(\frac{2i-1}{4}\pi\right) + W_1$$

$$X_2 = \sqrt{E} \sin\left(\frac{2i-1}{4}\pi\right) + W_2$$

MPSK:

$$S_k(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t + \frac{2\pi(k-1)}{M}\right), \quad k = 1, 2, \dots, M, \quad 0 \leq t \leq T.$$

pulse shape form

$$= \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t + \frac{2\pi(k-1)}{M}\right) \cdot g(t), \quad g(t) = \text{rect}\left(\frac{t}{T} - \frac{1}{2}\right)$$

Received signal with delay:

$$X(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi(f_c - \tau_c)t + \alpha_k\right) g(t - \tau_g) + W(t)$$

τ_c : carrier delay, cause carrier phase shift.

τ_g : group / envelope delay, symbol waveform overall shift.

Let $\theta = -2\pi f_c \tau_c$ and $\tau_g = \tau$.

$$\text{Then, } X(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t + \underbrace{\theta}_{\text{相位誤差}} + \alpha_k\right) \cdot g(t) + W(t), \quad \tau \leq t \leq T + \tau.$$

If receiver has f_c and τ , then the basis can be defined as

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t), \quad \phi_2(t) = -\sqrt{\frac{2}{T}} \sin(2\pi f_c t), \quad \tau \leq t \leq T + \tau.$$

Received sample is $x_i(\tau) = \int_{\tau}^{T+\tau} x(t) \phi_i(t) dt$, $i = 1, 2$.

Signal component is $s_i(a_k, \theta, \tau) = \int_{\tau}^{T+\tau} \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \theta + \alpha_k) \phi_i(t) dt$, $i = 1, 2$,

where $a_k = \{\cos \alpha_k, \sin \alpha_k\}$.

If $f_c = \frac{n_c}{T}$, $n_c \in \mathbb{N}$, then

$$s(a_k, \theta, \tau) = \begin{bmatrix} s_1(a_k, \theta, \tau) \\ s_2(a_k, \theta, \tau) \end{bmatrix} = \begin{pmatrix} \sqrt{E} \cos(\theta + \alpha_k) \\ \sqrt{E} \sin(\theta + \alpha_k) \end{pmatrix}$$

carrier phase error θ rotate the constellation

Received sample model:

$$x_k(\tau) = s(a_k, \theta, \tau) + w, \text{ where } w = [w_1, w_2]^T \text{ and } w_i \sim N(0, \frac{N_0}{2})$$

$\rightarrow X|a_k, \theta, \tau \sim 2D$ Gaussian with mean $s(a_k, \theta, \tau)$.

Conditional PDF:

$$f_X(x|a_k, \theta, \tau) = \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} \|x_k(\tau) - s(a_k, \theta, \tau)\|^2\right) \text{ 在已知 } a_k, \theta, \tau \text{ 時得到 } x \text{ 的 PDF}$$

$\hookrightarrow x_1, x_2$ independent, joint PDF is $f_{x_1} \cdot f_{x_2}$

Likelihood function:

$$L(a_k, \theta, \tau) = \frac{f_X(x|a_k, \theta, \tau)}{f_X(x|s(a_k, \theta, \tau) = 0)} \rightarrow \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} \|x(\tau)\|^2\right)$$

$$= \exp\left(\frac{2}{N_0} x_k^T(\tau) s(a_k, \theta, \tau) - \frac{1}{N_0} \|s(a_k, \theta, \tau)\|^2\right)$$

For MPSK, $\|s(a_k, \theta, \tau)\|^2$ is a constant E since all the symbol energy are the same.

$$\text{Thus, } L(a_k, \theta, \tau) = \exp\left(\frac{2}{N_0} x_k^T(\tau) s(a_k, \theta, \tau)\right).$$

重要的是 $x_k^T \cdot s$

If a sequence of symbols $a = [a_0, a_1, \dots, a_{L-1}]$ is sent, then

$$l(a_k, \theta, \tau) = \log \prod_{k=0}^{L-1} L(a_k, \theta, \tau) = \frac{2}{N_0} \sum_{k=0}^{L-1} x_k^T s(a_k, \theta, \tau).$$

$$\frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} (x_{1,k} \cos(\theta + \hat{\alpha}_k) + x_{2,k} \sin(\theta + \hat{\alpha}_k))$$

$$= \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} \operatorname{Re} \{ \tilde{a}_k^* \tilde{x}_k e^{-j\theta} \}$$

$\frac{\partial}{\partial \theta} \downarrow$

$$= \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} \operatorname{Im} \{ \tilde{a}_k^* \tilde{x}_k e^{-j\theta} \}$$

Decision - Directed phase recovery:

$$\hat{x}_k(\theta) = s(\hat{a}_k, \theta, \tau) = \sqrt{E} \begin{pmatrix} \cos(\theta + \hat{\alpha}_k) \\ \sin(\theta + \hat{\alpha}_k) \end{pmatrix}$$

已知 $\hat{a}_k, \tau$ 求估計 θ .

Phase log-likelihood:

$$\tilde{x}_k = x_{1,k} + j x_{2,k} \text{ and } \tilde{a}_k = \cos \hat{\alpha}_k + j \sin \hat{\alpha}_k$$

$$\operatorname{Re} \{ \tilde{a}_k^* \tilde{x}_k e^{-j\theta} \}$$

$$\text{Let } x_k = [x_{1,k}, x_{2,k}]^T.$$

$$\text{Then, } l(\theta) = \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} (x_{1,k} \cos(\theta + \hat{\alpha}_k) + x_{2,k} \sin(\theta + \hat{\alpha}_k))$$

這是在比較 received point 跟「旋轉了 θ 的 estimated constellation point」有多像。

$$= \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} [(x_{1,k} \cos \hat{\alpha}_k + x_{2,k} \sin \hat{\alpha}_k) \cos \theta - (x_{1,k} \sin \hat{\alpha}_k - x_{2,k} \cos \hat{\alpha}_k) \sin \theta]$$

偏微求極值:

$$0 = \frac{\partial}{\partial \theta} l(\theta) = -\frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} [(x_{1,k} \cos \hat{\alpha}_k + x_{2,k} \sin \hat{\alpha}_k) \sin \theta + (x_{1,k} \sin \hat{\alpha}_k - x_{2,k} \cos \hat{\alpha}_k) \cos \theta]$$

Complex form:

$$\text{Let } \tilde{x}_k = x_{1,k} + j x_{2,k} \text{ and } \tilde{a}_k = \exp(j \hat{\alpha}_k) = \cos \hat{\alpha}_k + j \sin \hat{\alpha}_k.$$

$$\text{Then, } \operatorname{Re} \{ \tilde{a}_k^* \tilde{x}_k \} = x_{1,k} \cos \hat{\alpha}_k + x_{2,k} \sin \hat{\alpha}_k \text{ (received sample 在 estimated symbol 上的投影)}$$

$$\operatorname{Im} \{ \tilde{a}_k^* \tilde{x}_k \} = -x_{1,k} \sin \hat{\alpha}_k + x_{2,k} \cos \hat{\alpha}_k \text{ (received sample 和 estimated symbol 的垂直誤差 } \Rightarrow \text{ phase error 來源)}$$

$$\frac{\partial}{\partial \theta} l(\theta) = \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} \operatorname{Re} \{ \tilde{a}_k^* \tilde{x}_k \} \operatorname{Im} \{ e^{-j\theta} \} + \operatorname{Im} \{ \tilde{a}_k^* \tilde{x}_k \} \operatorname{Re} \{ e^{-j\theta} \}$$

$$= \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L-1} \operatorname{Im} \{ \tilde{a}_k^* \tilde{x}_k e^{-j\theta} \}$$

ML phase estimator :

$$\Delta 0 = \frac{\partial}{\partial \theta} \lambda(\theta) = \frac{2\sqrt{E}}{N_0} \sum_{k=0}^{L_0-1} \text{Im} \{ \tilde{a}_k^* \tilde{x}_k e^{-j\theta} \}$$

By method of gradient ascent,

$$\begin{aligned} \hat{\theta}[n+1] &= \hat{\theta}[n] + \gamma' \frac{\partial \lambda(\theta)}{\partial \theta} \bigg|_{\hat{\theta}[n]} \\ &= \hat{\theta}[n] + \gamma' \text{Im} \{ \tilde{a}_k^* \tilde{x}_k e^{-j\hat{\theta}[n]} \} \\ &\triangleq e[n] \end{aligned}$$

上面的已有 $\tilde{a}_k, \tau, \tau_c$, 要估 θ

下面的只有 τ_c , 要估 τ

Nondata - Aided symbol timing recovery :

$$\text{We have } L(a_k, \theta, \tau) = \exp \left(\frac{2}{N_0} \chi_k^T(\tau) s(a_k, \theta, \tau) \right).$$

因為 a_k, θ 未知, 要先消滅

Averaging θ :

$$\begin{aligned} L(a_k, \tau) &= \frac{1}{2\pi} \int_0^{2\pi} \exp \left(\frac{2}{N_0} \chi_k^T(\tau) s(a_k, \theta, \tau) \right) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \exp \left(\frac{2\sqrt{E}}{N_0} \text{Re} \{ \tilde{a}_k^* \tilde{x}_k(\tau) e^{-j\theta} \} \right) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \exp \left(\frac{2\sqrt{E}}{N_0} |\tilde{a}_k| |\tilde{x}_k(\tau)| \cos(\arg(\tilde{x}_k(\tau)) - \arg(\tilde{a}_k) - \theta) \right) d\theta \end{aligned}$$

$$\text{Let } \phi = \arg(\tilde{x}_k(\tau)) - \arg(\tilde{a}_k) - \theta.$$

$$\text{Modified Bessel function of the first kind : } I_0(x) = \frac{1}{2\pi} \int_0^{2\pi} \exp(x \cos \phi) d\phi$$

$$\text{Then, } L(a_k, \tau) = I_0 \left(\frac{2\sqrt{E}}{N_0} |\tilde{a}_k| |\tilde{x}_k(\tau)| \right)$$

$$\text{For MPSK, } |\tilde{a}_k| = 1, \text{ so } L(a_k, \tau) = I_0 \left(\frac{2\sqrt{E}}{N_0} |\tilde{x}_k(\tau)| \right) \triangleq L(\tau)$$

$\Rightarrow$ 對 θ 平均後, 也不依賴 a_k 了!

If observing L_0 numbers of independent symbols,

then,

$$\begin{aligned} \lambda_{av}(\tau) &= \prod_{k=0}^{L_0-1} L(a_k, \tau) \\ &= \prod_{k=0}^{L_0-1} I_0 \left(\frac{2\sqrt{E}}{N_0} |\tilde{x}_k(\tau)| \right) \end{aligned}$$

$$\lambda_{av}(\tau) = \sum_{k=0}^{L_0-1} \log \left(I_0 \left(\frac{2\sqrt{E}}{N_0} |\tilde{x}_k(\tau)| \right) \right)$$

$$\text{Power series of Bessel function : } I_0(x) = \sum_{m=0}^{\infty} \frac{(\frac{x}{2})^{2m}}{(m!)^2}$$

$$\text{If } x \rightarrow 0, I_0(x) \doteq 1 + \frac{x^2}{4} \text{ and then } \log(I_0(x)) \doteq \frac{x^2}{4}$$

$$\Rightarrow \lambda_{av}(\tau) \doteq \sum_{k=0}^{L_0-1} \frac{E}{N_0} |\tilde{x}_k(\tau)|^2$$

The goal becomes find the $\hat{\tau}$ s.t. $\hat{\tau} = \underset{\tau}{\text{argmax}} \sum_k |\tilde{x}(\tau)|^2$

$$\begin{aligned} 0 &= \frac{\partial}{\partial \tau} \lambda_{av}(\tau) = \frac{E}{N_0} \sum_{k=0}^{L_0-1} \frac{\partial}{\partial \tau} |\tilde{x}_k(\tau)|^2 \\ &\quad \downarrow \frac{\partial}{\partial \tau} |\tilde{x}_k(\tau)|^2 = 2 \text{Re} \{ \tilde{x}_k^*(\tau) \tilde{x}_k'(\tau) \} \\ &= \frac{2E}{N_0} \sum_{k=0}^{L_0-1} \text{Re} \{ \tilde{x}_k^*(\tau) \tilde{x}_k'(\tau) \} \end{aligned}$$

Using gradient ascent method :

$$\hat{t}_{n+1} = \hat{t}_n + \gamma' \frac{\partial}{\partial t} \log(\tau) \Big|_{\hat{t}_n} = \hat{t}_n + \gamma' \operatorname{Re} \{ \tilde{x}_n^*(\tau) \tilde{x}_n'(\tau) \} \Big|_{\hat{t}_n}$$

$$\text{Define } \tilde{x}_n(\tau) = \tilde{x}(n\tau + \hat{t}_n) \quad \text{and} \quad \tilde{x}_n'(\tau) = \tilde{x}'(n\tau + \hat{t}_n)$$

$$\tilde{x}_n'(\tau) \stackrel{\text{middle}}{=} \frac{1}{T} \left(\tilde{x}(n\tau + \frac{T}{2} + \hat{t}_{n+1}) - \tilde{x}(n\tau - \frac{T}{2} + \hat{t}_{n-1}) \right)$$

$\hat{t}_n$ $\hat{t}_{n-1}$

$$\Rightarrow e(n) = T \cdot \operatorname{Re} \{ \tilde{x}_n^*(\tau) \tilde{x}_n'(\tau) \}$$

$$\stackrel{\text{early, late 的差乘上 middle 的共轭判断新 sample point 要往前 or 後}}{=} \operatorname{Re} \{ \tilde{x}^*(n\tau + \hat{t}_n) [\tilde{x}(n\tau + \frac{T}{2} + \hat{t}_n) - \tilde{x}(n\tau - \frac{T}{2} + \hat{t}_{n-1})] \}$$

(early, late 的差乘上 middle 的共轭判断新 sample point 要往前 or 後)

Thus, we get $\hat{t}_{n+1} = \hat{t}_n + \gamma e(n)$

Direct-Sequence Spread Spectrum

Transmitted signal :

$$m(t) = c(t) b(t).$$

$c(t)$ is wideband pseudo-noise signal. (通常取 $c(t) = \pm 1$)

$b(t)$ is narrowband message signal.

↳ 做資料展寬

Received signal :

$$r(t) = m(t) + i(t) = c(t)b(t) + i(t), \text{ where } i(t) \text{ is additive interference.}$$

De-spreading :

$$z(t) = c(t)r(t) = c^2(t)b(t) + c(t)i(t) = b(t) + c(t)i(t).$$

↳ 資料解展頻, 干擾展寬

DSSS with coherent BPSK :

$$s(t) = \pm \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

$$x(t) = c(t)s(t) = \pm \sqrt{\frac{2E_b}{T_b}} c(t) \cos(2\pi f_c t)$$

實際流程 : data $\rightarrow$ PN spreading $\rightarrow$ BPSK modulation) 因為是相乘, 交換時分析不影響

分析 : data $\rightarrow$ BPSK modulation $\rightarrow$ PN spreading

$$y(t) = x(t) + j(t) \quad (\text{先 BPSK modulation})$$

$$c(t)y(t) = c^2(t)s(t) + c(t)j(t) = s(t) + c(t)j(t) \quad (\text{和上面一樣})$$

Signal - Space Dimensionality :

Assume that there are N chips in a bit : $N = \frac{T_b}{T_c} \rightarrow \begin{matrix} \text{bit duration} \\ \text{chip duration} \end{matrix}$ (chip interval)

Basis functions of every chip :

$$\phi_k(\tau) = \begin{cases} \sqrt{\frac{1}{T_c}} \cos(2\pi f_c \tau) & , kT_c \leq \tau \leq (k+1)T_c \\ 0 & , \text{o.w.} \end{cases}$$

quadrature

$$\tilde{\phi}_k(\tau) = \begin{cases} \sqrt{\frac{1}{T_c}} \sin(2\pi f_c \tau) & , kT_c \leq \tau \leq (k+1)T_c \\ 0 & , \text{o.w.} \end{cases}$$

$$k = 0, 1, \dots, N-1.$$

Let $\{c_0, c_1, \dots, c_{N-1}\}$ denote the PN sequence with $c_i = \pm 1, \forall i$.

$$\text{Then, } x(t) = \pm \sqrt{\frac{2E_b}{T_b}} c(t) \cos(2\pi f_c t) = \pm \sqrt{\frac{E_b}{N}} \sum_{k=0}^{N-1} c_k \phi_k(\tau)$$

↳ N -dimensional since we use $\phi_1, \dots, \phi_N$

Jamming signal :

$$\text{Let } j_k = \int_0^{T_c} j(t) \phi_k(\tau) d\tau \quad \text{and} \quad \tilde{j}_k = \int_0^{T_c} j(t) \tilde{\phi}_k(\tau) d\tau$$

$$\text{Then, } j(t) = \sum_{k=0}^{N-1} j_k \phi_k(t) + \sum_{k=0}^{N-1} \tilde{j}_k \tilde{\phi}_k(t). \quad (N+N = 2N - \text{dimensional})$$

Jammer power :

$$J \triangleq \frac{1}{T_b} \int_0^{T_b} j^2(t) dt = \frac{1}{T_b} \sum_{k=0}^{N-1} j_k^2 + \sum_{k=0}^{N-1} \tilde{j}_k^2 = \frac{2}{T_b} \sum_{k=0}^{N-1} j_k^2$$

$$\Rightarrow \sum_{k=0}^{N-1} j_k^2 = \frac{J T_b}{2}$$

↳ 最佳策略是把能量平均在 sine, cosine.

After receiver de-spreading, $u(t) = s(t) + c(t)j(t)$.

Define $v = \sqrt{\frac{2}{T_b}} \int_0^{T_b} u(t) \cos(2\pi f_c t) dt = V_s + V_{cj}$.

Then, signal component $V_s = \sqrt{\frac{2}{T_b}} \int_0^{T_b} s(t) \cos(2\pi f_c t) dt$
 $= \pm \frac{2\sqrt{E_b}}{T_b} \int_0^{T_b} \cos^2(2\pi f_c t) dt$
 $= \pm \sqrt{E_b}$

$$s(t) = \pm \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

Also, jamming component $V_{cj} = \sqrt{\frac{2}{T_b}} \int_0^{T_b} c(t)j(t) \cos(2\pi f_c t) dt$
 $= \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k j_k$

$$= \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} \int_{kT_c}^{(k+1)T_c} C_k j(t) \cos(2\pi f_c t) dt$$

$$= \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k \cdot \int_{kT_c}^{(k+1)T_c} j(t) \cos(2\pi f_c t) dt$$

$$= \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k \cdot \int_{kT_c}^{(k+1)T_c} j(t) \sqrt{\frac{T_c}{2}} \phi_k(t) dt$$

$$\downarrow \phi_k(t) = \sqrt{\frac{2}{T_c}} \cos(2\pi f_c t)$$

$$= \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k \cdot j_k$$

Mean and Variance of V_{cj} :

Consider PN code as an i.i.d binary sequence,

then $P(C_k = 1) = P(C_k = -1) = \frac{1}{2}$

Let $V_{cj} = \sqrt{\frac{T_c}{T_b}} \sum_{k=0}^{N-1} C_k j_k$.

For fixed j_k , we have

$$E[C_k j_k | j_k] = P(C_k = 1) j_k + P(C_k = -1) (-j_k)$$

$$= \frac{1}{2} j_k - \frac{1}{2} j_k$$

$$= 0$$

$$\text{Var}[V_{cj} | j] = \frac{T_c}{T_b} \sum_{k=0}^{N-1} j_k^2 = \frac{T_c}{T_b} \cdot \frac{JT_b}{2} = \frac{JT_c}{2}$$

Output SNR:

$$\text{SNR}_o = \frac{V_s^2}{\text{Var}[V_{cj} | j]} = \frac{E_b}{JT_c/2} = \frac{2E_b}{JT_c}$$

Input SNR:

$$\text{SNR}_i = \frac{P}{J} = \frac{E_b}{J \cdot T_b} \quad \left(P = \frac{E_b}{T_b} \quad \text{power} = \frac{\text{energy}}{\text{time}} \right)$$

Define the process gain $PG = \frac{T_b}{T_c} = N$.

Then, $\text{SNR}_o = 2PG \text{SNR}_i$.

$$\Rightarrow 10 \log(\text{SNR}_o) = 10 \log(\text{SNR}_i) + 3 + \frac{10 \log(PG)}{\text{展頻增益}}$$

同 coherent detection 的

BER:

Detector output $V = \pm \sqrt{E_b} + V_{cj}$.

The error is $P(V > 0 | 0) = P(V_{cj} > \sqrt{E_b})$

By CLT, $V_{cj} \sim N(0, \frac{JT_c}{2})$

Since BER of BPSK is $\frac{1}{2} \text{erfc}(\sqrt{\frac{E_b}{N_0}})$ if $n \sim N(0, \frac{N_0}{2})$

Thus, $P_e = \frac{1}{2} \text{erfc}(\sqrt{\frac{E_b}{JT_c}})$ ($\because \frac{N_0}{2} = \frac{JT_c}{2}$ here)

Anti-jam Characteristic:

Since $N_b = JT_c$ and $E_b = PT_b$, $\frac{E_b}{N_b} = \frac{PT_b}{JT_c} = \frac{P}{J} PG$

Thus, jamming margin $\frac{J}{P} = \frac{PG}{E_b/N_b}$

To achieve the specific BER, the minimum require is $(\frac{E_b}{N_b})_{\min}$.

Then, (jamming margin)_{dB} = (PG)_{dB} - $10 \log(\frac{E_b}{N_b})_{\min}$

Frequency-Hop Spread Spectrum

如果 DSSS 的 PG 還不夠大，可以改用 frequency hopping。

概念是：

不要固定在同一個 carrier frequency 傳，而是讓 carrier frequency 依照 PN sequence 在多個頻率之間跳動。

因此 jammer 若想干擾你，必須干擾整個 hopping bandwidth。

這會迫使 jammer 把功率分散到很寬的頻帶，單位頻率上的干擾強度下降。

FH / MFSK :

$$M = 2^k, \text{ where } k \text{ is the bit number of symbol.}$$

Slow FH :

$$R_s = n R_h, \quad n \in \mathbb{N}. \quad (\text{several symbol per hop})$$

symbol rate hop rate

Fast FH :

$$R_h = n R_s, \quad n \in \mathbb{N}. \quad (\text{several hop per symbol})$$

★ For a k -bit PN code, there are 2^k FHs.

Chip Rate of FH / MFSK :

$$R_c = \max(R_h, R_s)$$

For Slow FH / MFSK :

$$R_c = R_s = \frac{R_h}{k} \geq R_h, \quad \text{where } k \text{ is the bit number of symbol.}$$

22.2 Slow FH 為什麼仍保持 orthogonality ?

MFSK 的 tone spacing 是 symbol rate 的整數倍，至少可以取：

$$\frac{R_s}{2}$$

因此不同 tone 之間仍然 orthogonal。

所以 slow FH/MFSK 的 symbol error 可以近似用 MFSK 在 AWGN 下的錯誤率分析。

The equivalent AWGN of jammer in FH / MFSK :

Jammer 把功率均勻分散到 FH bandwidth W_c

$$\Rightarrow \text{Equivalent noise spectral density is } N_0 = \frac{J}{W_c}.$$

$$\text{Symbol energy is } E = \frac{P}{R_s}$$

$$\text{Thus, } \frac{E}{N_0} = \frac{P/R_s}{J/W_c} = \frac{P \cdot W_c}{J \cdot R_s} \quad (\text{只要 } W_c/R_s \text{ 很大, effective SNR } \uparrow)$$

△ Processing gain of FH/MFSK :

$$\text{Output bandwidth of MFSK is } B \approx 2^{k-1} R_s$$

$$\text{Define the processing gain is } PG \triangleq \frac{W_c}{B}$$

$$\text{If PN code has } k \text{ bits, then } W_c = 2^k B$$

$$\text{Thus, } PG = 2^k.$$

$$\Rightarrow (PG)_{dB} = 10 \log 2^k \approx 3k \text{ dB}$$

Jamming margin of FH/MFSK:

$$\frac{E}{N_0} = \frac{P}{J} \cdot \frac{W_c}{R_s}$$

$$\Rightarrow \frac{J}{P} = \frac{W_c \cdot N_0}{E \cdot R_s}$$

$$= \frac{2^k \cdot 2^{K-1} R_s \cdot N_0}{E \cdot R_s}$$

$$= 2^k \cdot 2^{K-1} \cdot \frac{N_0}{E}$$

$$\Rightarrow 10 \log\left(\frac{J}{P}\right) = 3k + 3(K-1) - 10 \log\left(\frac{E}{N_0}\right)_{\min}$$

$$= (PG)_{\text{dB}} + 3(K-1) - 10 \log\left(\frac{E}{N_0}\right)_{\min}$$

Fast FH/MFSK detection:

1. Majority vote approach.

2. Maximum likelihood approach.

Error-control coding

Block code and code rate:

(n, k) block code:

Input is $m = [m_0, m_1, \dots, m_{k-1}]$

Output is $c = [c_0, c_1, \dots, c_{n-1}]$

$\hookrightarrow n-k$ parity bits.

Code rate: $r = \frac{k}{n}$ (效率和保護力或反比)

Linear block code:

$$c_1, c_2 \in B_2^n \Rightarrow c_1 + c_2 \in B_2^n$$

$\hookrightarrow$ mod 2 code book.

Remark: If we have generator matrix G or parity-check matrix H , then we can operate encoding or decoding.

Systematic linear block code:

$c = [b : m]$ $\hookrightarrow$ parity $\hookrightarrow$ message

$$b = [b_0, b_1, \dots, b_{n-k-1}]$$

$$m = [m_0, m_1, \dots, m_{k-1}]$$

b_i is the linear combination of m : $b_i = \sum_{n=0}^{k-1} \alpha_{n,i} m_n$, $\forall i$.

$$\Rightarrow b = mP, \quad P = \begin{pmatrix} P_{0,0} & P_{0,1} & \dots & P_{0,n-k-1} \\ P_{1,0} & P_{1,1} & \dots & P_{1,n-k-1} \\ \vdots & \vdots & \ddots & \vdots \\ P_{k-1,0} & P_{k-1,1} & \dots & P_{k-1,n-k-1} \end{pmatrix}$$

Thus, $c = m[P : I_k]$, i.e. $G = [P : I_k]$ is generator matrix.

$$\Rightarrow c = mG$$

Generator Matrix G and Parity-check Matrix H :

$$c = mG, \quad G = [P : I_k]$$

$$H = [I_{n-k} : P^T]$$

If c is a legal codeword vector, $cH^T = 0$.

$$cH^T = mGH^T$$

$$= m[P : I_k] \begin{bmatrix} I_{n-k} \\ P \end{bmatrix}$$

$$I_n \tilde{z}_i^0 = m(P + P)$$

$$= 0$$

Syndrome :

$$r = c + e \quad \text{接收的}$$

$$\text{For } e, e_i = \begin{cases} 1, & \text{if } i\text{-th is wrong} \\ 0, & \text{if } i\text{-th is correct} \end{cases}$$

$$s = rH^T = eH^T$$

↓
全法 code $cH^T = 0$

Hamming distance :

Two codewords c_1, c_2 .

$d(c_1, c_2)$ is Hamming distance

In binary code, $d(c_1, c_2) = w(c_1 + c_2)$
↳ Hamming weight

$$c_1 = 1100 \Rightarrow d(c_1, c_2) = 2$$

$$c_2 = 1001$$

$$c_1 + c_2 = 0101 \Rightarrow w(c_1 + c_2) = 2$$

Hamming weight :

$w(c)$ is defined as the numbers of 1 in c .

$$\text{e.g. } w(1011000) = 3.$$

Minimum distance :

$$d_{\min} = \min_{i \neq j} d(c_i, c_j)$$

For linear code c_i, c_j , we have $c_i + c_j = c_k$.

$$\text{Thus, } d(c_i, c_j) = w(c_i + c_j) = w(c_k)$$

$$\Rightarrow d_{\min} = \min_{c \neq 0} w(c).$$

Minimum distance decoding :

Received input : $r = c_i + e$.

$$\hat{c} = \underset{c_i}{\operatorname{argmin}} d(r, c_i)$$

To guarantee that t errors can be corrected, we need $d_{\min} \geq 2t + 1$

$$\Rightarrow t \leq \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor$$

推導直覺 :

每個 codeword 周圍畫半徑 t 的球。

若兩個 codeword 距離至少 $2t + 1$, 則兩個球不會重疊。

所以只要錯誤數 $\leq t$, 收到的點一定離原本 codeword 最近。

Standard array and coset leader :

Standard array :

$$\begin{array}{ccccccc} c_1 = 0 & c_2 & c_3 & \cdots & c_{2^k} \\ e_1 & e_1 + c_2 & & \cdots & e_1 + c_{2^k} \\ \vdots & \vdots & & & \vdots \\ e_{n-k} & e_{n-k} + c_2 & \cdots & & e_{n-k} + c_{2^k} \end{array} \quad \rightarrow \text{each row is called a coset}$$

↑
coset leader

Decoding :

For each syndrome $s = rH^T$, find its coset leader e_i ,

and then $\hat{c} = r + e_i$

Hamming code :

It's a kind of linear block code.

For $m \geq 3$:

$$n = 2^m - 1$$

$$k = 2^m - m - 1 \quad \text{number of message bits}$$

$$n - k = m$$

↓
number of parity bits

(7, 4) Hamming code with $m = 3$:

$$n = 2^3 - 1 = 7$$

$$k = 2^3 - 3 - 1 = 4$$

$$m = 7 - 4 = 3$$

The generator matrix is $G = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} = [P : I_4]$

The parity check matrix is $H = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$
 $= [I_{n-k} : P^T]$

TABLE 10.1 Code words of a (7, 4) Hamming code

Message Word	Code Word	Weight of Code Word	Message Word	Code Word	Weight of Code Word
0000	0000000	0	1000	1101000	3
0001	1010001	3	1001	0111001	4
0010	1110010	4	1010	0011010	3
0011	0100011	3	1011	1001011	4
0100	0110100	3	1100	1011100	4
0101	1100101	4	1101	0001101	3
0110	1000110	3	1110	0101110	4
0111	0010111	4	1111	1111111	7

The minimum distance of (7, 4) Hamming code is $d_{\min} = 3$

$$t = \left\lfloor \frac{3-1}{2} \right\rfloor = 1$$

$\Rightarrow$ It can only correct 1 error.

Ex:

■ Suppose $c = [1110010]$ is sent

■ And $r = [1100010]$

■ Then, $s = rH^T = [0 \ 0 \ 1]$, which corresponds to $e_3 = [0010000]$

■ $\Rightarrow$ Flip the 3rd bit of r , yielding $\hat{c} = [1110010] = c$

~~2x~~

4x

syndrome is $[0 \ 0 \ 1] \Rightarrow e_3 = [0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0]$ since $H = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$

$$\hat{c} = r + e_3 = [1110010]$$

Dual code :

(n, k) code with generator matrix G and parity-check matrix P

$\Downarrow$ Dual

$(n, n-k)$ code with generator matrix P and parity-check matrix G

Cyclic code :

Special form of linear block code and it satisfies two properties :

1. linearity

2. cyclic property : A codeword $c = [c_0, c_1, \dots, c_{n-1}]$ after cyclic shifting is still a codeword.

✳ It can use a polynomial to express.

Code polynomial :

$$c = [c_0, c_1, \dots, c_{n-1}]$$

$$\Rightarrow c(x) = c_0 + c_1x + c_2x^2 + \dots + c_{n-1}x^{n-1} \text{ where } c_i \in \{0, 1\} \ \forall i$$

In binary field, i -th bit cyclic shift equal to multiple x^i , i.e. $x^i c(x) \bmod (x^n + 1)$

Generator polynomial :

$$g(x) = 1 + g_1x + g_2x^2 + \dots + g_{n-k-1}x^{n-k-1} + x^{n-k}$$

For each legal code polynomial $c(x)$ can be expressed as $c(x) = a(x)g(x)$,

where $\deg a(x) \leq k-1$ and $g(x)$ must be the factor of $x^n + 1$.

Systematic cyclic code :

$$\text{message poly. : } m(x) = m_0 + m_1x + \dots + m_{k-1}x^{k-1}$$

$$\text{parity poly. : } b(x) = b_0 + b_1x + \dots + b_{n-k-1}x^{n-k-1}$$

$$\text{Then, } c(x) = b(x) + m(x)x^{n-k}$$

For a legal codeword $c(x)$, if we want to get the parity bits, then $b(x)$ is the remainder of $x^{n-k} m(x)$ divided by $g(x)$.

$$c(x) = a(x)g(x) \text{ and } c(x) = b(x) + m(x)x^{n-k}.$$

$$b(x) + m(x)x^{n-k} = a(x)g(x)$$

$$\frac{m(x)x^{n-k}}{g(x)} = a(x) + \frac{b(x)}{g(x)} \text{ 余数.}$$

$\Rightarrow$ 有 $m(x)$, $g(x)$ 就可推 $c(x)$.

Parity-check polynomial :

$$h(x) = 1 + h_1x + h_2x^2 + \dots + h_{n-k}x^{n-k} + x^n \text{ with } g(x)h(x) = x^n + 1$$

$$H = \begin{pmatrix} \text{Coeff. of } h(x) \\ \text{coeff. of } x \cdot h(x) \\ \vdots \\ \text{coeff. of } x^{n-k+1} \cdot h(x) \end{pmatrix}$$

Cyclic code syndrome :

$$r = [r_0, r_1, \dots, r_{n-1}]$$

$$r(x) = r_0 + r_1x + \dots + r_{n-1}x^{n-1}.$$

$$r(x) = g(x)q(x) + \frac{s(x)}{\text{remainder}}$$

$s(x)$ is syndrome polynomial and $\deg s(x) \leq n-k-1$

Since $r(x) = c(x) + e(x)$ and $c(x) = a(x)g(x)$,

$$\begin{aligned} e(x) &= c(x) + r(x) \\ &= a(x)g(x) + g(x)q(x) + r(x) \\ &= (a(x) + q(x))g(x) + r(x) \end{aligned}$$

Thus, $s(x) \equiv e(x) \pmod{g(x)}$.

If $\deg e(x) \leq n-k-1$, then $s(x) = e(x)$ and $\hat{c}(x) = r(x) + s(x)$.

Cyclic code of (7,4) Hamming code :

$$n = 7$$

$$k = 4$$

$$m = 3$$

$$\text{For } 1001, m(x) = 1 + x^3$$

$$x^{n-k}m(x) = x^3(1+x^3) = x^3 + x^6$$

$$x^7 + 1 = (1+x)(1+x+x^3)(1+x+x^3)$$

$$\text{Let } g(x) = 1+x+x^3$$

$$\frac{x^{n-k}m(x)}{g(x)} = \frac{x^3+x^6}{1+x+x^3} = a(x) + x + x^2$$

$$\text{Thus, } b(x) = x + x^2$$

$$c(x) = b(x) + x^{n-k}m(x) = x + x^2 + x^3 + x^6$$

$$\Rightarrow \text{bit vector is } \underline{0111001} \text{ parity}$$

Maximum-length code (MLC) :

MLC is the dual code of Hamming code.

(7, 3) MLC :

Dual code of (7, 4) Hamming code .

$$g(x) = \frac{1+x^7}{1+x+x^3} = 1+x+x^2+x^4$$

Convolution code

message length is L , memory is M , rate $\frac{1}{n}$

$\Rightarrow$ output length is $n(L+M)$.

$$\text{Code rate } r = \frac{L}{n(L+M)}.$$

Constraint length

$$K = M+1.$$

$\hookrightarrow$ 1 个 input bit 影响 的长度 (M memory + 自己一个)

Generator polynomial :

$$g^{(i)}(D) = g_0^{(i)} + g_1^{(i)}D + \dots + g_M^{(i)}D^M.$$

message sequence polynomial is $m(D)$

$$\Rightarrow C^{(i)}(D) = m(D) \cdot g^{(i)}(D)$$

e.g.

$$g^{(1)}(D) = 1 + D + D^2$$

$$g^{(2)}(D) = 1 + D^2$$

message is [10011]

$$\Rightarrow m(D) = 1 + D^3 + D^4$$

$$\begin{aligned} C^{(1)}(D) &= (1 + D + D^2)(1 + D^3 + D^4) \\ &= 1 + D + D^3 + D^3 + D^4 + D^4 + D^5 + D^6 \end{aligned}$$

$$\begin{aligned} C^{(2)}(D) &= (1 + D^2)(1 + D^3 + D^4) \\ &= 1 + D + D^2 + D^3 + D^4 + D^5 + D^6 \end{aligned}$$

ML decoding :

In BSC,

$$p(r_i | c_i) = \begin{cases} 1-p & , r_i = c_i \\ p & , r_i \neq c_i \end{cases}$$

$$\text{Then, } p(r|c) = \prod_{i=1}^N p(r_i | c_i)$$

$$\Rightarrow \log p(r|c) = \sum_{i=1}^N \log p(r_i | c_i)$$

如果 r, c 有 d 个位置不同 $\rightarrow p^d \cdot (1-p)^{N-d}$

$$\begin{aligned} \Rightarrow \log p(r|c) &= d \log p + (N-d) \log (1-p) \\ &= d \log \frac{p}{1-p} + N \log (1-p) \end{aligned}$$

Since $p < \frac{1}{2}$, $\frac{p}{1-p} < 1$ and then $\log \frac{p}{1-p} < 0$

Thus, $d \downarrow$, $\log p(r|c) \uparrow$.

$$\Leftrightarrow \hat{c} = \arg \min_c d(r, c)$$

Free distance d_{free} :

Block code 用 d_{min} .

$$\begin{aligned} \text{Convolution code 用 } d_{\text{free}} &= \min_{c_i \neq c_j} d_H(c_i, c_j) \\ &= \min_{c \neq 0} w_H(c) \end{aligned}$$

可修正 t errors 的条件: $d_{\text{free}} > 2t$

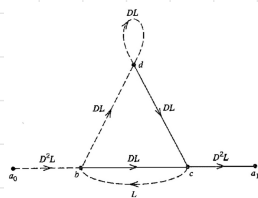
$$\Leftrightarrow t \leq \left\lfloor \frac{d_{\text{free}} - 1}{2} \right\rfloor$$

Distance transfer function :

Every branch marks $D^w L$.

w : The output Hamming weight of the branch.

L : one step.



$$b = D^2 L a_0 + L c$$

$$c = D L b + D L d$$

$$d = D L b + D L d$$

$$a_1 = D^2 L c$$

$$\Rightarrow T(D, L) \triangleq \frac{a_1}{a_0}$$

$$= \frac{D^2 L^3}{1 - D L (1 + L)}$$

$$= D^2 L^3 \sum_{i=0}^{\infty} (D L (1 + L))^i$$

Let $L = 1$.

$$\text{Then, } T(D, 1) = \underbrace{D^5}_{\substack{\downarrow \\ \text{1 條} \\ d=5}} + \underbrace{2D^6}_{\substack{\downarrow \\ \text{2 條} \\ d=6}} + \underbrace{4D^7}_{\substack{\downarrow \\ \text{4 條} \\ d=7}} + \dots$$

Asymptotic Coding Gain :

Uncoded binary PSK :

$$P_b \sim \exp\left(-\frac{E_b}{N_0}\right)$$

After convolution code :

$$P_b \sim \exp\left(-\frac{d_{free} \cdot r \cdot E_b}{2 N_0}\right)$$

$$\Rightarrow ACG = 10 \log_{10} \left(\frac{d_{free} \cdot r}{2} \right) \text{ dB}$$